

# CS 2770: Pitt CRC Tutorial

Prof. Nils Murrugarra- Llerena

## [Setup]

- Clone this Github repo: [https://github.com/nineil-pitt/tutorial\\_pitt\\_CRC](https://github.com/nineil-pitt/tutorial_pitt_CRC) locally in your computer. You can use Git command line or [Github Desktop](#).

## [Jupyter Notebooks]

We are following this [website documentation](#) to connect to Jupyter Notebooks on Pitt CRC.

1. Login on [Jupyter Hub](#) with your Pitt Credentials
2. Select your Partition with GPU (e.g. TEACH – Nvidia GTX 1080 GPU) and account: cs2770\_2026s

## Server Options

### JupyterHub Session Configuration

Select Partition:

TEACH - 6 CPUs - 45GB

Select Virtual Environment:

base

Select Modules to Load:

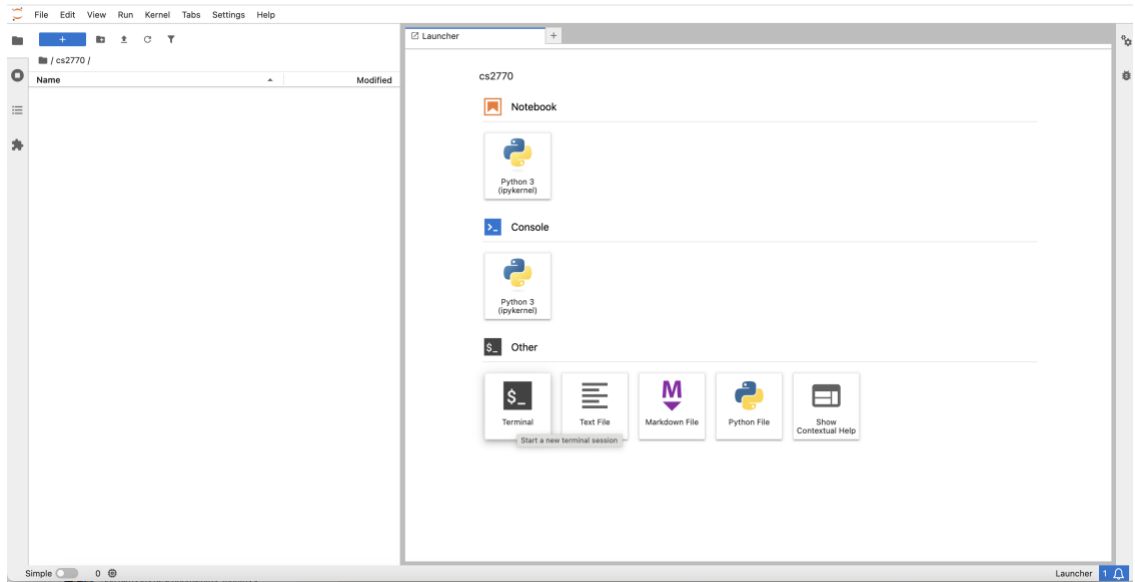
Amber 2024  
Cuda 12.3  
OpenJDK 21.0.2

Hold Ctrl/Cmd to select multiple modules

Account:

Start

3. Once logged in, select “terminal”



4. Verify that you have a machine with GPU using command: `nvidia-smi`

```
nem177@teach-gpu-n24:~$ nvidia-smi
Thu Jan 22 21:09:29 2026
```

| NVIDIA-SMI 575.57.08 |                            |      |               | Driver Version: 575.57.08 |                     | CUDA Version: 12.9 |             |
|----------------------|----------------------------|------|---------------|---------------------------|---------------------|--------------------|-------------|
| GPU                  | Name                       | Perf | Persistence-M | Bus-Id                    | Disp.A              | Volatile           | Uncorr. ECC |
| Fan                  | Temp                       |      | Pwr:Usage/Cap |                           | Memory-Usage        | GPU-Util           | Compute M.  |
|                      |                            |      |               |                           |                     |                    | MIG M.      |
| 0                    | NVIDIA GeForce GTX 1080 Ti | P8   | On 9W / 250W  | 00000000:3B:00.0          | Off 3MiB / 11264MiB | 0%                 | N/A         |
| 23%                  | 32C                        |      |               |                           |                     |                    | Default     |
|                      |                            |      |               |                           |                     |                    | N/A         |

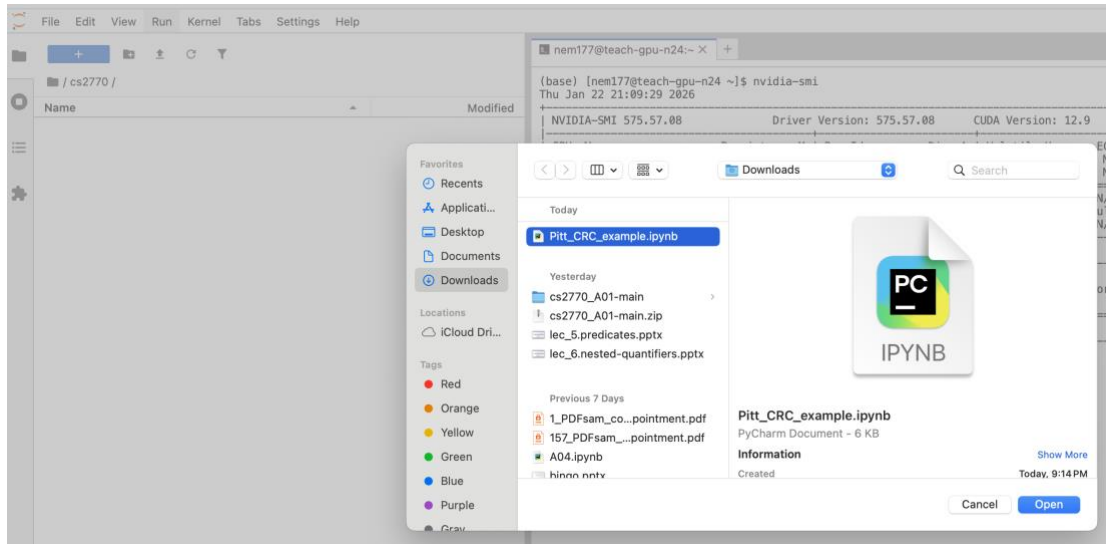
  

```
Processes:
```

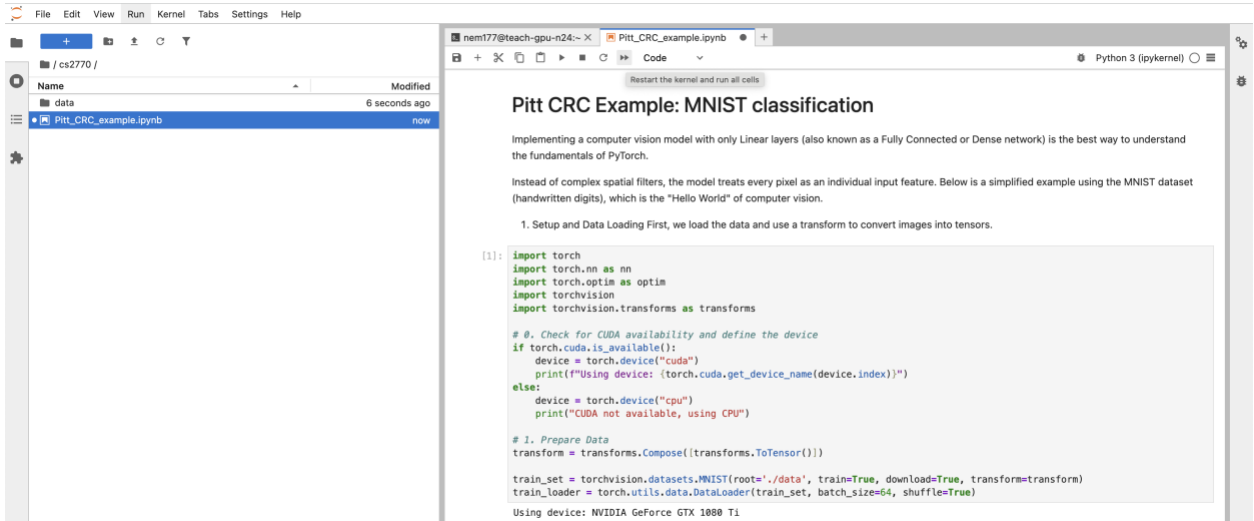
| GPU                        | GI | CI | PID | Type | Process name | GPU Memory Usage |
|----------------------------|----|----|-----|------|--------------|------------------|
| ID                         | ID |    |     |      |              |                  |
| No running processes found |    |    |     |      |              |                  |

```
(base) [nem177@teach-gpu-n24 ~]$
```

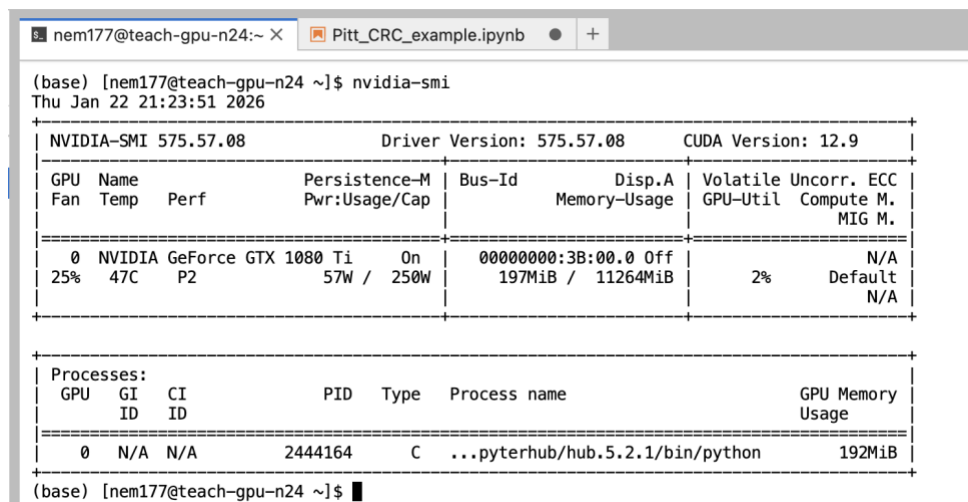
5. Upload Jupyter Notebook Pitt\_CRC\_example.ipynb with button 



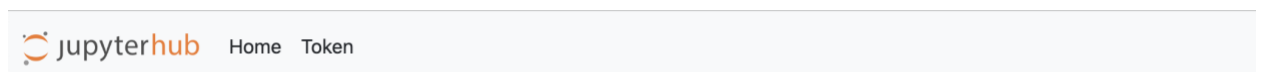
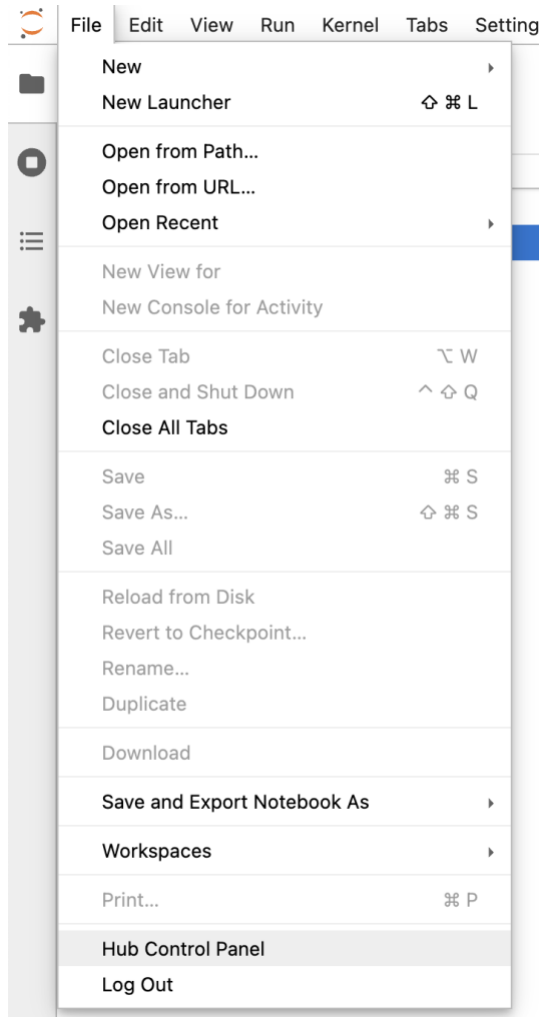
## 6. Open Jupyter Notebook and Execute with button ▶



## 7. Open a new terminal window, and monitor the use of GPU



## 8. Logout using File, Hub Control Panel, Stop My Server



### [Jupyter Ondemand]

We are following this [website documentation](#) to connect to Jupyter OnDemand on Pitt CRC.

1. Login to [Open OnDemand](#) with your Pitt Credentials

OPEN OnDemand

OPEN OnDemand

Log in with your HPC username and password.

Username

Password

Log in to Open OnDemand

- Select on “all available apps” and search for “Jupyter on GPU”

Open OnDemand Apps Files Jobs Clusters Interactive Apps Bioimage Apps Genomics Apps My Interactive Sessions

OPEN OnDemand

OnDemand provides an integrated, single access point for all of your HPC resources.

Pinned Apps A featured subset of all available apps



Jupyter

System Installed App



RStudio Server 2025

System Installed App



Code Server

System Installed App



matlab

System Installed App



COMSOL Multiphysics

System Installed App

Open OnDemand Apps Files Jobs Clusters Interactive Apps Bioimage Apps Genomics Apps My Interactive Sessions Help

Home / All Apps

Show 10 entries

Search: jupyter

| Name                                                                                               | Category         | Sub Category  |
|----------------------------------------------------------------------------------------------------|------------------|---------------|
|  Jupyter        | Interactive Apps | Servers       |
|  Jupyter on gpu | Interactive Apps | Deep Learning |

Showing 1 to 2 of 2 entries (filtered from 52 total entries)

Previous

1

Next

- Select your configuration (e.g. # of hours, # of cores, GPU type, # of GPUs, etc) and add account to “cs2770\_2026s” and click on launch.

## Bioimage Apps

### GUIs

 CellProfiler

 Fiji

 Napari

 QuPath

 ilastik

### fMRI

 3D-Slicer

 AFNI

 FSL

 Freeview

 ITK-SNAP

 MRICron

 MRtrix3

 dsi\_studio

## Jupyter on gpu

This app will launch a [Jupyter](#) server using [Python](#) on the [gpu cluster].

☒ Use JupyterLab instead of Jupyter Notebook?

JupyterLab is the next generation of Jupyter with an IDE-like experience, and is completely compatible with existing Jupyter Notebooks.

### Python version

pytorch 2.5.1 python 3.11 cuda 12.4

This defines the version of python you want to load.

### Cuda version

cuda 12.9.0

This defines the version of cuda you want to load. **You must choose None if you are using PyTorch and the associated PyTorch-CUDA packages.**

### Number of hours

1

### Number of cores

1

Number of CPU cores [1-128] on node (8 GB per core unless requesting whole node). Leave blank if requesting single core.

dsi\_studio

Genomics Apps

GUIs

IGV on htc

Nextflow

nf-core pipelines

Proteomics

FragPipe

Servers

IGV webapp

Shiny Apps

Statistics and Machine Learning

single-cell

Azimuth v0.4.3

ChromScape 1.8.0

Omics Playground 2.8.0

cellxgene

singlecellTK 2.18.0

Interactive Apps

Apps

R Shiny

Number of CPU cores [1-128] on node (8 GB per core unless requesting whole node). Leave blank if requesting single core.

CPU Memory (GB) (optional)

Amount of CPU memory to allocate

GPU type

a100

constraint

amd,40g

Number of gpu cards

1

Number of GPU cards [1-4] on node. Leave blank if requesting single card.

Account

cs2770\_2026s

The allocation you would like to use for SLURM.

Reservation

You can leave this blank if there is **no** reservation.

☒ I would like to receive an email when the session starts

Launch

\* The Jupyter on gpu session data for this session can be accessed under the [data root directory](#).

- Wait while the system assigns you resources, you will receive an email.

Session was successfully created.

Home / My Interactive Sessions

Bioimage Apps

GUIs

CellProfiler

Fiji

Napari

QuPath

ilastik

fMRI

Jupyter on gpu (1582321)

Queued

Created at: 2026-01-23 15:51:47 EST

Time Requested: 1 hour

Session ID: 8ed4dcdc-4a22-4293-a5ef-13b8b3e8621c

Please be patient as your job currently sits in queue. The wait time depends on the number of cores as well as time requested.

Delete

- Once you received the email, you can connect using Open OnDemand interface. Click on “Connect to Jupyter”

**Jupyter on gpu (1582321)** 1 node | 12 cores | Running

**Host:** `>_ gpu-n38.crc.pitt.edu` Delete

**Created at:** 2026-01-23 15:51:47 EST

**Time Remaining:** 57 minutes

**Session ID:** `8ed4dcdc-4a22-4293-a5ef-13b8b3e8621c`

Connect to Jupyter

- Then, you can do same steps as our previous section (working with Jupyter Notebooks)

File Edit View Run Kernel Tabs Settings Help

/ cs2770 /

| Name                   | Modified      |
|------------------------|---------------|
| data                   | 19 hours ago  |
| Pitt_CRC_example.ipynb | 19 hours ago  |
| run.sh                 | 2 minutes ago |

nem177@gpu-n38:~/cs2770

(base) [nem177@gpu-n38 cs2770]\$ nvidia-smi  
Fri Jan 23 16:36:31 2026

| NVIDIA-SMI |                       | 575.57.08     |                  | Driver Version: 575.57.08 |                 | CUDA Version: 12.9   |                   |
|------------|-----------------------|---------------|------------------|---------------------------|-----------------|----------------------|-------------------|
| GPU        | Name                  | Persistence-M | Bus-Id           | Disp.A                    | Memory-Usage    | Volatile Uncorr. ECC |                   |
| Fan        | Temp                  | Perf          | Pwr:Usage/Cap    |                           |                 | GPU-Util             | Compute M. MIG M. |
| 0          | NVIDIA A100-PCIE-40GB | On            | 00000000:47:00:0 | Off                       | 0MiB / 40960MiB | 0%                   | Default Disabled  |
| N/A        | 31C                   | P0            | 35W / 250W       |                           |                 |                      |                   |

Processes:

| GPU                        | GI | CI | PID | Type | Process name | GPU Memory Usage |
|----------------------------|----|----|-----|------|--------------|------------------|
| ID                         | ID | ID |     |      |              |                  |
| No running processes found |    |    |     |      |              |                  |

(base) [nem177@gpu-n38 cs2770]\$

- Once you finish, you can shut down this machine.

File Edit View Run Kernel Tabs Settings

New

New Launcher

Open from Path...

Open from URL...

Open Recent

New View for

New Console for Activity

Close Tab

Shutdown Terminal

Close All Tabs

Save

Save As...

Save All

Reload from Disk

Revert to Checkpoint...

Rename...

Duplicate

Download

Save and Export Notebook As

Workspaces

Print...

Log Out

Shut Down



## [Command Line: Interactive Job]

We are following this [website documentation](#) from Pitt CRC.

1. Connect via ssh with command: “ssh [<yourpittID>@h2p.crc.pitt.edu](#)” in your terminal for linux or Mac. If you are using windows, please try [MobaXterm](#) or [Putty](#).

```
((base) nem177@Nilss-MacBook-Pro ~ % ssh nem177@h2p.crc.pitt.edu
nem177@h2p.crc.pitt.edu's password:
#####

Welcome to h2p.crc.pitt.edu!

Documentation can be found at https://crc-pages.pitt.edu/user-manual/

-----

IMPORTANT REMINDERS

Don't run jobs on login nodes! Use interactive jobs: `crc-interactive --help`

Slurm is separated into 'clusters', e.g. if `scancel <jobnum>` doesn't work
try `crc-scancel <jobnum>`. Try `crc-sinfo` to see all clusters.

-----

#####
Last login: Fri Jan 23 16:01:11 2026 from 10.25.197.87
(base) [nem177@login0 ~]$
```

2. Clone this Github repo: [https://github.com/nineil-pitt/tutorial\\_pitt\\_CRC](https://github.com/nineil-pitt/tutorial_pitt_CRC) on Pitt CRC with command “git clone [https://github.com/nineil-pitt/tutorial\\_pitt\\_CRC.git](https://github.com/nineil-pitt/tutorial_pitt_CRC.git)” and move to tutorial\_pitt\_CRC/ with command “cd tutorial\_pitt\_CRC”
3. Observe state of GPU resources with command: “sinfo --clusters=gpu”. Partition l40s seems available.

```
((base) [nem177@login1 ~]$ sinfo --clusters=gpu
CLUSTER: gpu
PARTITION      AVAIL  TIMELIMIT  NODES  STATE NODELIST
a100*           up    infinite    12    mix  gpu-n[33-44]
a100_multi      up    infinite     1  down* gpu-n53
a100_multi      up    infinite     9  alloc gpu-n[45-52,54]
preempt         up    infinite     1  down* gpu-n53
preempt         up    infinite    16    mix  gpu-n[58-59,61-73],mems-n0
preempt         up    infinite    13  alloc gpu-n[45-52,54-57,60]
a100_nvlink     up    infinite     4    mix  gpu-n[29-32]
a100_nvlink     up    infinite     1  alloc gpu-n28
a100_nvlink_multi up    infinite     4    mix  gpu-n[29-32]
a100_nvlink_multi up    infinite     1  alloc gpu-n28
l40s            up    infinite    15    mix  gpu-n[58-59,61-73]
l40s            up    infinite     4  alloc gpu-n[55-57,60]
```

4. Request interactive node: “srun --clusters=gpu --gres=gpu:1 --partition=l40s --nodes=1 --time=01:00:00 --pty bash”

```

((base) [nem177@login0 cs2770]$ srun --clusters=gpu --gres=gpu:1 --partition=l40s --nodes=1 --time=01:00:00 --pty bash
srun: job 1582339 queued and waiting for resources
srun: job 1582339 has been allocated resources
((base) [nem177@gpu-n55 cs2770]$ nvidia-smi
Fri Jan 23 17:40:00 2026

```

| NVIDIA-SMI 575.57.08 |             |      | Driver Version: 575.57.08 |                  |              | CUDA Version: 12.9 |            |     |
|----------------------|-------------|------|---------------------------|------------------|--------------|--------------------|------------|-----|
| GPU                  | Name        | Perf | Persistence-M             | Bus-Id           | Disp.A       | Volatile           | Uncorr.    | ECC |
| Fan                  | Temp        |      | Pwr:Usage/Cap             |                  | Memory-Usage | GPU-Util           | Compute M. |     |
|                      |             |      |                           |                  |              |                    | MIG M.     |     |
| 0                    | NVIDIA L40S |      | On                        | 00000000:E1:00:0 | Off          |                    |            | 0   |
| N/A                  | 22C         | P8   | 32W / 350W                | 0MiB / 46068MiB  |              | 0%                 | Default    | N/A |

| Processes:                 |    |    |     |      |              |            |  |
|----------------------------|----|----|-----|------|--------------|------------|--|
| GPU                        | GI | CI | PID | Type | Process name | GPU Memory |  |
| ID                         | ID |    |     |      |              | Usage      |  |
| No running processes found |    |    |     |      |              |            |  |

## 5. Execute your code

### a. Jupyter Notebook Option

#### i. Execute Jupyter Notebook with command: “papermill

Pitt\_CRC\_example.ipynb out.ipynb”

```

((base) [nem177@gpu-n55 cs2770]$ papermill Pitt_CRC_example.ipynb out.ipynb
Input Notebook: Pitt_CRC_example.ipynb
Output Notebook: out.ipynb
Executing: 0%|
Executing notebook with kernel: python3
Executing: 100%|

```

#### ii. The generated output is stored in out.ipynb notebook

### b. Python Code Option

#### i. Execute Python code with command: “python Pitt\_CRC\_example.py”

```

((base) [nem177@gpu-n55 cs2770]$ python Pitt_CRC_example.py
Using device: NVIDIA L40S
Epoch 0
Batch 0: Loss = 2.2995
Batch 500: Loss = 1.0016
Epoch 1
Batch 0: Loss = 0.5489
Batch 500: Loss = 0.5350
Epoch 2
Batch 0: Loss = 0.3408
Batch 500: Loss = 0.3494
Epoch 3
Batch 0: Loss = 0.3059
Batch 500: Loss = 0.2231
Epoch 4
Batch 0: Loss = 0.4119
Batch 500: Loss = 0.2242
Epoch 5
Batch 0: Loss = 0.2699
Batch 500: Loss = 0.4217
Epoch 6
Batch 0: Loss = 0.4221
Batch 500: Loss = 0.3276
Epoch 7
Batch 0: Loss = 0.3909
Batch 500: Loss = 0.3017
Epoch 8
Batch 0: Loss = 0.3090
Batch 500: Loss = 0.2182
Epoch 9
Batch 0: Loss = 0.2445
Batch 500: Loss = 0.3315

```

## [Command Line: Batch Job]

We are following this [website documentation](#) from Pitt CRC.

1. Connect via ssh with command: “ssh <pittID>@h2p.crc.pitt.edu” in your terminal for linux or Mac. If you are using windows, please try [MobaXterm](#) or [Putty](#).

```

[(base) nem177@Nilss-MacBook-Pro ~ % ssh nem177@h2p.crc.pitt.edu
nem177@h2p.crc.pitt.edu's password:
#####

Welcome to h2p.crc.pitt.edu!

Documentation can be found at https://crc-pages.pitt.edu/user-manual/

-----

IMPORTANT REMINDERS

Don't run jobs on login nodes! Use interactive jobs: `crc-interactive --help`

Slurm is separated into 'clusters', e.g. if `scancel <jobnum>` doesn't work
try `crc-scancel <jobnum>`. Try `crc-sinfo` to see all clusters.

-----

#####
Last login: Fri Jan 23 16:01:11 2026 from 10.25.197.87
(base) [nem177@login0 ~]$

```

2. Update your Pitt email in run.sh and run\_python.sh scripts

3. In Pitt CRC, execute your code

a. Jupyter Notebook Code

i. Type “sbatch run.sh” command.

```

[(base) [nem177@gpu-n55 cs2770]$ sbatch run.sh
Submitted batch job 1582341 on cluster gpu

```

ii. While executing, run a command to the state of your job using “queue --clusters=gpu --partition=l40s --user=<yourPittID>”

```

[(base) [nem177@login1 cs2770]$ squeue --clusters=gpu --partition=l40s --user=nem177
CLUSTER: gpu

```

| JOBID   | PARTITION | NAME     | USER   | ST | TIME  | NODES | ODELIST(REASON) |
|---------|-----------|----------|--------|----|-------|-------|-----------------|
| 1582339 | l40s      | bash     | nem177 | R  | 37:27 | 1     | gpu-n55         |
| 1582341 | l40s      | test_277 | nem177 | R  | 0:09  | 1     | gpu-n61         |

b. Python Code

i. Type “sbatch run\_python.sh” command.

```

[(base) [nem177@login0 cs2770]$ sbatch run_python.sh
Submitted batch job 1582364 on cluster gpu

```

ii. After the command finishes, you can inspect files: code\_gpu-1582364.err and code\_gpu-1582364.out with cat command

```

[(base) [nem177@login0 cs2770]$ cat code_gpu-1582364.err
Lmod Warning:

The following dependent module(s) are not currently loaded: libxml2/2.10.3
(required by: cuda/12.1.1)

-----

The following have been reloaded with a version change:
1) libiconv/1.17-b47av4d => libiconv/1.17-oygrtg5
2) libxml2/2.10.3 => libxml2/2.13.5-ftxdjfd
3) xz/5.4.1 => xz/5.4.6-anmq7m4
4) zlib-ng/2.1.5-g64y7ia => zlib-ng/2.2.3-sp4tqsg

[notice] A new release of pip is available: 24.0 -> 25.3
[notice] To update, run: pip3 install --upgrade pip

```

```

[base] [nem177@login0 cs2770]$ cat code_gpu-1582364.out
Defaulting to user installation because normal site-packages is not writeable
Requirement already satisfied: Pillow in /ihome/nllexerena/nem177/.local/lib/python3.11/site-packages (12.1.0)
Executing in node: gpu-n61.crc.pitt.edu
Date: Fri Jan 23 06:51:59 PM EST 2026
Working Directory: /ihome/nllexerena/nem177/cs2770
GPU information:
Fri Jan 23 18:51:59 2026

```

| NVIDIA-SMI 575.57.08 |             |      | Driver Version: 575.57.08      |                  | CUDA Version: 12.9     |                                                       |
|----------------------|-------------|------|--------------------------------|------------------|------------------------|-------------------------------------------------------|
| GPU                  | Name        | Perf | Persistence-M<br>Pwr:Usage/Cap | Bus-Id           | Disp.A<br>Memory-Usage | Volatile Uncorr. ECC<br>GPU-Util Compute M.<br>MIG M. |
| 0                    | NVIDIA L40S | P8   | On<br>33W / 350W               | 00000000:4A:00:0 | Off<br>0MiB / 46068MiB | 0<br>Default<br>N/A                                   |

```


```

| GPU                        | GI | CI | PID | Type | Process name | GPU Memory<br>Usage |
|----------------------------|----|----|-----|------|--------------|---------------------|
| No running processes found |    |    |     |      |              |                     |

```


```

Initializing code...

Input: /ihome/nllexerena/nem177/cs2770//Pitt\_CRC\_example.py

Using device: NVIDIA L40S

Epoch 0

Batch 0: Loss = 2.3222

Batch 500: Loss = 0.9446

Epoch 1

Batch 0: Loss = 0.5379

Batch 500: Loss = 0.4536

Epoch 2

Batch 0: Loss = 0.5789

Batch 500: Loss = 0.3883

Epoch 3

Batch 0: Loss = 0.2842

Batch 500: Loss = 0.2730

Epoch 4

Batch 0: Loss = 0.2942

Batch 500: Loss = 0.3258

Epoch 5

Batch 0: Loss = 0.4418

Batch 500: Loss = 0.1103

Epoch 6

Batch 0: Loss = 0.3798

Batch 500: Loss = 0.2339

Epoch 7

Batch 0: Loss = 0.5312

Batch 500: Loss = 0.3298

Epoch 8

Batch 0: Loss = 0.3859

Batch 500: Loss = 0.3477

Epoch 9

Batch 0: Loss = 0.2134

Batch 500: Loss = 0.3049

Code executed correctly!

Job finished on: Fri Jan 23 06:52:27 PM EST 2026

4. Finally, generate errors for our examples and run again
  - a. For Jupyter Notebook: run.sh script and Python code: run\_python.sh script, comment the line “# unset LD\_LIBRARY\_PATH” and submit the job.
  - b. What are the responses from Slurm?
    - i. Do you receive an email?
    - ii. Can you inspect error and output files?
    - iii. What about output Jupyter notebook?

## [FAQ]

Q: How to transfer files to Pitt CRC?

A: To transfer files to Pitt CRC, you can follow [this documentation](#).

Q: How to send a program in background in an interactive session?

A: You can use [screen](#) Linux command.

Q: Where can I find more documentation of slurm?

A: You can review [slurm documentation](#) and [Pitt CRC website](#).

Q: How can I cancel a job on Slurm?

A: Use command “scancel **<jobID>**”. See [documentation](#).

Q: How can I know the state of a job on Slurm?

A: Use command “scontrol --clusters=gpu show job **<jobID>**”. See [documentation](#).

Q: How to create a conda virtual environment?

A: Follow the [guide](#) from Pitt CRC.

Q: What modules/packages/programs are available to load?

A: Please, review them from this [website](#).

Q: Where can I find more help from Pitt CRC?

A: Please, review their website <https://crc.pitt.edu/>